



Assessing **Critical Thinking** Components in **Romanian Secondary School** **Textbooks**: A Data Mining Approach to the ROTEX Corpus

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Background and motivation

- Critical thinking is essential for analyzing, evaluating, and synthesizing information (Ennis, 1985).
 - Skills include argument evaluation, metacognition, and questioning source credibility.
 - Challenge: Generative AI may reduce reflective thinking (Lee et al., 2025).
 - Textbooks play a key role in promoting critical thinking but are often not evaluated at scale.
 - Traditional analysis: Manual coding (Halpern, 1998).
- ✓ Emerging solution: NLP enables large-scale assessment of: argumentative density, discourse coherence, syntactic complexity



Research questions

RQ1: To what extent do Romanian language textbooks at the secondary level include tasks that support higher-order cognitive processes, as defined by Bloom's Taxonomy?

RQ2: How are these tasks distributed across grades and chapters, and do they reflect a coherent pedagogical progression?

RQ3: Which textbook series demonstrate the most effective use of instructional tasks for promoting critical thinking, based on verb-level cognitive classification?

Related work

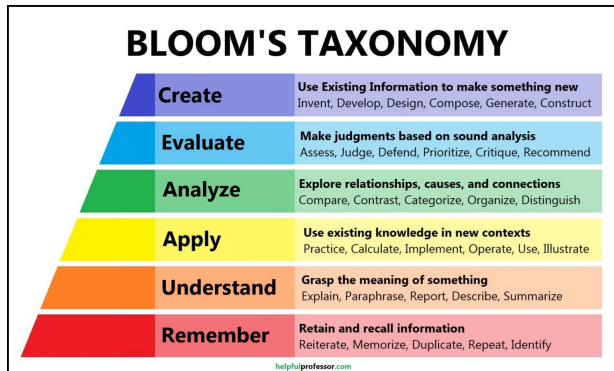




Thinking and critical thinking

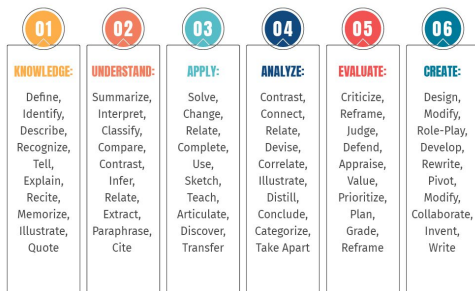
- Thinking is active and purposeful: used to understand, question, and revise beliefs (Dewey, 2022; Nosich, 2005).
- Reflective thinking involves careful evaluation of beliefs and evidence.
- Critical thinking adds judgment: questions assumptions, evaluates arguments, identifies knowledge gaps
- Involves both fast (intuitive) and slow (deliberate) processes (Kahneman, 2011).
- Seen as the most analytical and self-aware form of thinking.
- Bloom's Taxonomy helps structure these skills in education.

Bloom's Taxonomy and linguistic research



- ✓ Oravițan et al. analyzed Romanian textbook tasks using verb patterns and found an overuse of high-level “create” tasks compared to mid-level skills like “analyze” (Oravițan et al., 2023).
- ✓ Graves examined university writing assignments and showed an uneven distribution of cognitive demands across Bloom’s levels, highlighting gaps in progression (Graves, 2017).

Textbook design

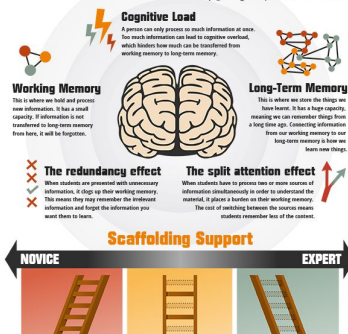


- ✓ Many textbooks focus heavily on lower-order thinking, limiting student growth in critical skills (Miyazaki, 2024; Riazzi & Mosalanejad, 2010).
- ✓ A scaffolded design based on Bloom’s Taxonomy supports deeper learning and should guide textbook development (Anderson & Krathwohl, 2001; Mizbani et al., 2023).

Multitasking and critical thinking

COGNITIVE LOAD THEORY: KEY TERMS

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- ✓ Overloading tasks with multiple cognitive demands can hinder critical thinking by overwhelming working memory and encouraging superficial engagement (Sweller, 1988; Torrance, 2007).
- ✓ Critical thinking develops best through focused, conceptually clear tasks, not multitasking, allowing students to deeply engage with one cognitive process at a time (Paul & Elder, 2007; Perkins, 2008).



ROTEX corpus

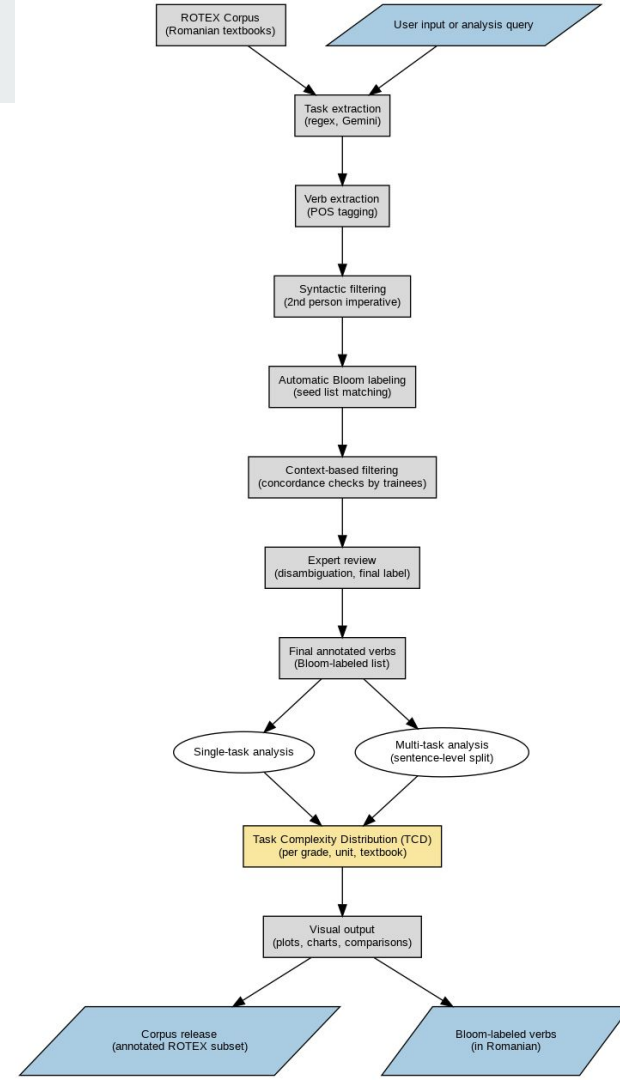
- The analysis draws on the ROLAT subset of the ROTES corpus (Chițez et al., 2024), which includes Romanian Language and Literature textbooks currently in use in secondary schools.
- Among publishers, only ArtKlett offers a complete series spanning grades 5 to 8, making longitudinal analysis across levels limited for other textbook series.

Textbook	5 th grade	6 th grade	7 th grade	8 th grade
ArtKlett	82,249	80,312	95,968	108,918
Booklet	63,416	93,031	-	-
Corint	58,719	85,836	-	93,214
Litera	55,191	67,895	77,288	-
Intuitext	71,279	78,211	85,143	-
CD	47,618	51,801	71,273	-
Press				
EDP	-	57,096	64,126	-
Paralela	-	88,567	99,002	-
45				
Ars	-	-	62,527	-
Libri				
Aramis	-	-	75,734	66,294
Total	1,880,708			

Table 1: ROTES sub-corpus size (no. words)

Annotation

- Bloom-Level Verb Annotation
- Annotated 434 task-related verbs from the ROTEX corpus with Bloom's Taxonomy labels.
- Used a 4-step pipeline: Task extraction (regex + Gemini prompting);= Verb extraction (spaCy POS tagging); Filtering for instructional verbs (2nd person); Bloom-level labeling (based on seed lists + expert review)
- Context-based annotation ensured accuracy: verbs were labeled within full task context.
- Most frequent levels: Understand (112), Apply (84), Analyze (86), Evaluate (79).
- Dataset available on GitHub





Results

Task Complexity Distribution (TCD) per textbook series

- Most publishers show **limited cognitive progression across grades**.
- ArtKlett*: 43% of tasks are lower-order (Remember, Understand), even in grade 8.
- CD Press*: 36% are Understand-level; only 12% are Create-level tasks.
- Intuitext*: Best scaffolding; clear growth in Analyze, Evaluate, and Create tasks by grade 8.
- Booklet*, *Paralela45*: Diverse task types but lack upward trajectory.
- Corint*, *Litera*, *Aramis*, *ArsLibri*: Remain static; minimal higher-order tasks.
- Overall, **55% of all textbook tasks in ROTEX (10,150 out of 18,518) remain at lower-order levels.**

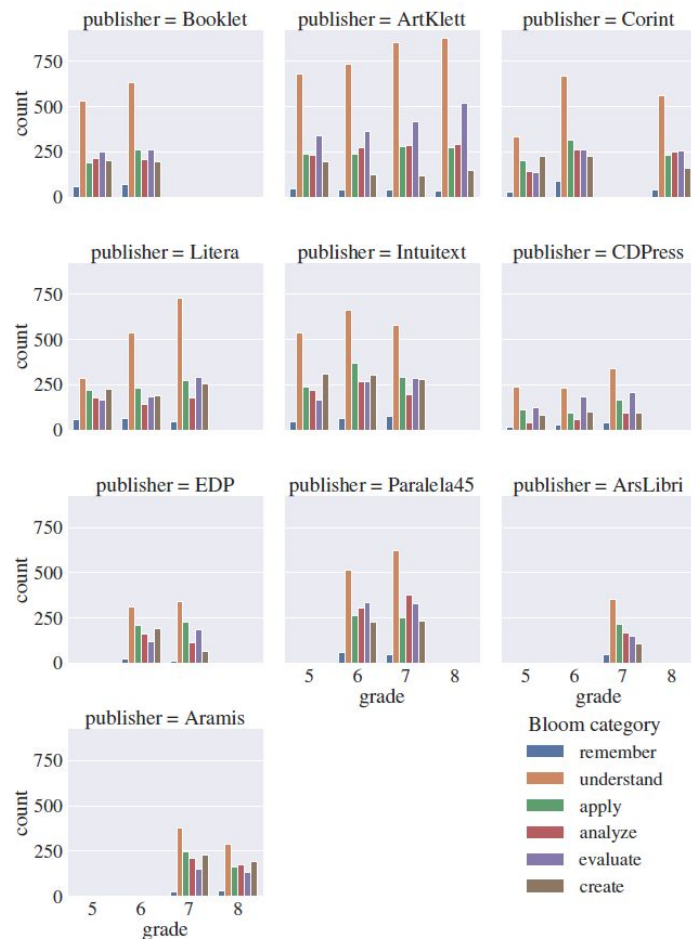


Figure 1: Bloom category counts by publisher and grade



Task Complexity Distribution (TCD) per learning unit within textbooks

- *Intuitext* shows clear cognitive progression across chapters, from Understand to Analyze, Evaluate, and Create, especially in grades 7-8.
- *CDPress* also displays a gradual shift toward higher-order tasks, though more lightweight in structure.
- *ArtKlett* includes higher-order tasks throughout but lacks sequencing or variation, leading to a dense, static design.
- *Booklet*, *Litera*, *Corint*, *Paralela45*, *Aramis*: flat distributions, mostly Remember/Understand, with minimal progression.
- Tasks often mix Bloom levels (e.g., “Justify your choice” = Evaluate + Create), revealing instructional inconsistency.
- Conclusion: Only a few textbook series embed higher-order thinking in a structured, chapter-based progression.



Task Complexity Distribution (TCD) within series across grades

- *Intuitext* is the only series with clear cognitive progression, more analyze, evaluate, and create tasks from grades 5 to 7.
- *CDPress* shows moderate growth in task complexity, especially in grades 6-7, but has limited overall coverage.
- *ArtKlett* distributes higher-order tasks evenly, resulting in cognitive density without structured scaffolding.
- *Booklet*, *Litera*, *Corint*, *Paralela45*, *Aramis*, *ArsLibri* show no significant change, tasks remain focused on lower-order thinking across all grades.
- Conclusion: Most series lack systematic cognitive progression across grade levels, which may hinder long-term critical thinking development.



Multi-task presence

- 26% of textbook tasks include multiple verbs linked to different Bloom levels.
- After sentence-level analysis, 15% remain as true multi-task exercises.
- These tasks often mix low- and high-order thinking (e.g., Understand + Create), creating ambiguity.
- Example (*ArtKlett*, Grade 5): “Select verbs” → Understand, “Write a story” → Create, “Reflect on friendship” → Evaluate → Mixed cognitive demands without clear focus.
- Most series, including *ArtKlett*, *Litera*, *Intuitext*, combine non-adjacent Bloom levels, lacking scaffolding logic.
- Suggests a lack of clear pedagogical structuring, which may confuse learning objectives.
- Only 15% of tasks show meaningful layering of Bloom levels; the rest reveal inconsistency, not progression.

Conclusions





How are the Romanian language secondary school textbooks in point of critical thinking competence support?

- **Limited cognitive progression:** Most Romanian language textbooks lack clear scaffolding across grades and chapters, with 55% of tasks targeting lower-order thinking (Remember, Understand).
- **Scattered higher-order tasks:** Only a few series (e.g., *Intuitext*, *CDPress*) show meaningful progression toward Analyze, Evaluate, and Create tasks.
- **Inconsistent task design:** 26% of exercises include multiple Bloom levels, but only 15% layer them in a pedagogically coherent way.
- **Structural gaps:** Mixed-task exercises often combine non-adjacent Bloom levels without guidance, reflecting weak alignment with competence development.
- **Need for redesign:** Findings highlight the need for textbook authors to implement clearer cognitive sequencing and task structuring aligned with Bloom's Taxonomy.



Limitations

- **Language-specific scope:** The analysis focuses solely on Romanian language textbooks, limiting cross-linguistic generalizability and comparison with other educational systems.
- **Verb-centric annotation:** Bloom-level labeling is based primarily on task verbs, which may oversimplify cognitive intent when tasks include complex or implicit instructional cues.

Mitigation:

To address ambiguity, all annotations were reviewed by trained student raters and verified by expert researchers, ensuring contextual accuracy and consistency.



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